CSC1058 USER INTERFACE DESIGN & IMPLEMENTATION

MODULE COORDINATOR & LECTURER
Hyowon Lee

TEACHING ASSISTANT
Haoyan Zhang

	Tuesday (3 – 4pm)	Wednesday (11am - 12pm)
Week 1 (7 Sep)	Introduction/ orientation	HCI background
Week 2 (14 Sep)	Design guidelines	Design guidelines
Week 3 (21 Sep)	Design guidelines	Design guidelines
Week 4 (28 Sep)	Design guidelines	Project brainstorming
Week 5 (5 Oct)	Guidelines recap	Guidelines recap
Week 6 (12 Oct)	Preventing negative impact	Usability evaluation
Week 7 (19 Oct)	Usability evaluation	Usability evaluation
Week 8 (26 Oct)	Usability evaluation	Usability evaluation
Week 9 (2 Nov)	Usability evaluation	Usability evaluation
Week 10 (9 Nov)	Design principles summary	Visualisation summary
Week 11 (16 Nov)	Project discussion	Project feedback
Week 12 (23 Nov)	Summary/ reflection	Summary/ reflection

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1

Design Principles

- Affordance
- Consistency
- Constraints
- ▶ **Feedback**
- Mapping
- Visibility



Eight Golden Rules

1. Strive for consistency
2. Cater to universal usability
3. Offer informative feedback
4. Design dialogs to yield closure
5. Prevent errors
6. Permit easy reversal of actions
7. Support internal locus of control
8. Reduce short-term memory load

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2

Design Princip

Affordance
Consistency
Constraints
▶ Feedback
Mapping
Visibility

DESIGN GUIDELINES

1. Be consistent
2. Simple and natural dialogue
   - Higher information/administration ratio
   - Graphic design concern (sharpening, grouping, layout, font choice, colour, emphasis)
3. Speak the user's language
   - Clear and user-oriented message, metaphor
4. Provide informative feedback
5. Reduce user's memory load
6. Provide short-cuts for frequent use
7. Good error messages
8. Prevent user-driven errors
9. Provide easy reversal of action

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Rules
consistency
iversal usability
rmative feedback
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t internal locus of control
e short-term memory load

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3

Design Principles

Affordance
Consistency
Constraints
Feedback
▶ Mapping
Visibility



Intuitive
Easy to use
User-friendly
:



Design by Taewon Hwang, Hyundae Asan, 2011

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4

Design Princi

- Affordance
- Consistency
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- ▶ **Mapping**
- Visibility





Design by Taewon Hwang, Hyundae Asan, 2011

Easy to use
User-friendly
:

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5

Design Principles

- Affordance
- Consistency
- Constraints
- Feedback
- Mapping
- ▶ **Visibility**

Making more visible:
- Functions
- Current state

Better access
Feeling of control
Less surprise



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6

Design Principles

Affordance
Consistency
Constraints
Feedback
Mapping
▶ Visibility



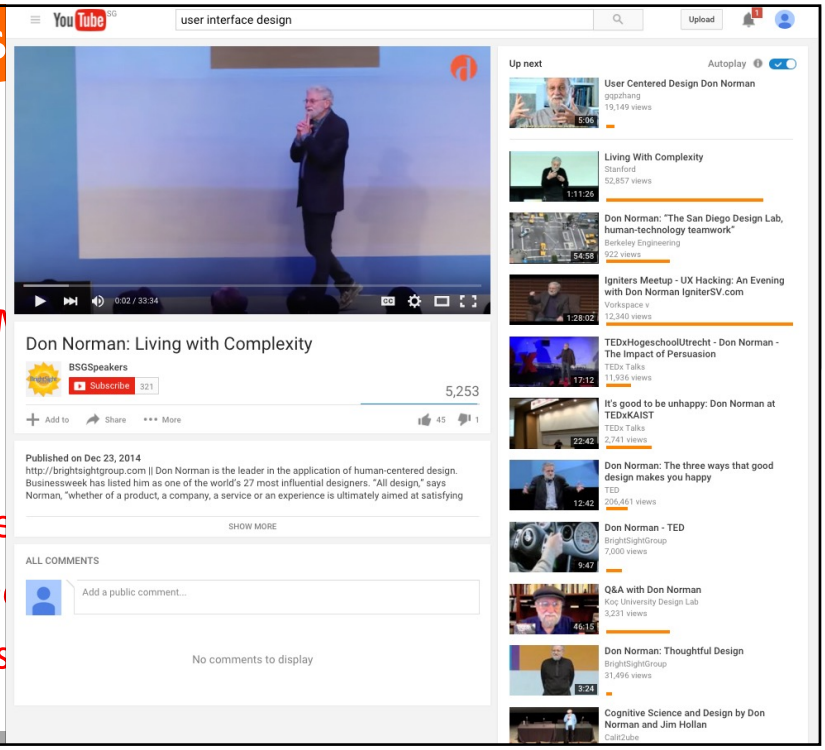
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• Fu
• Current state
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7

Design Principles

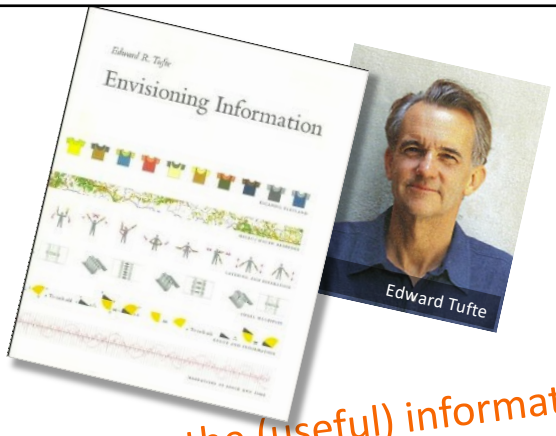
Affordance
Consistency
Constraints
Feedback
Mapping
▶ Visibility



8

Design Principles

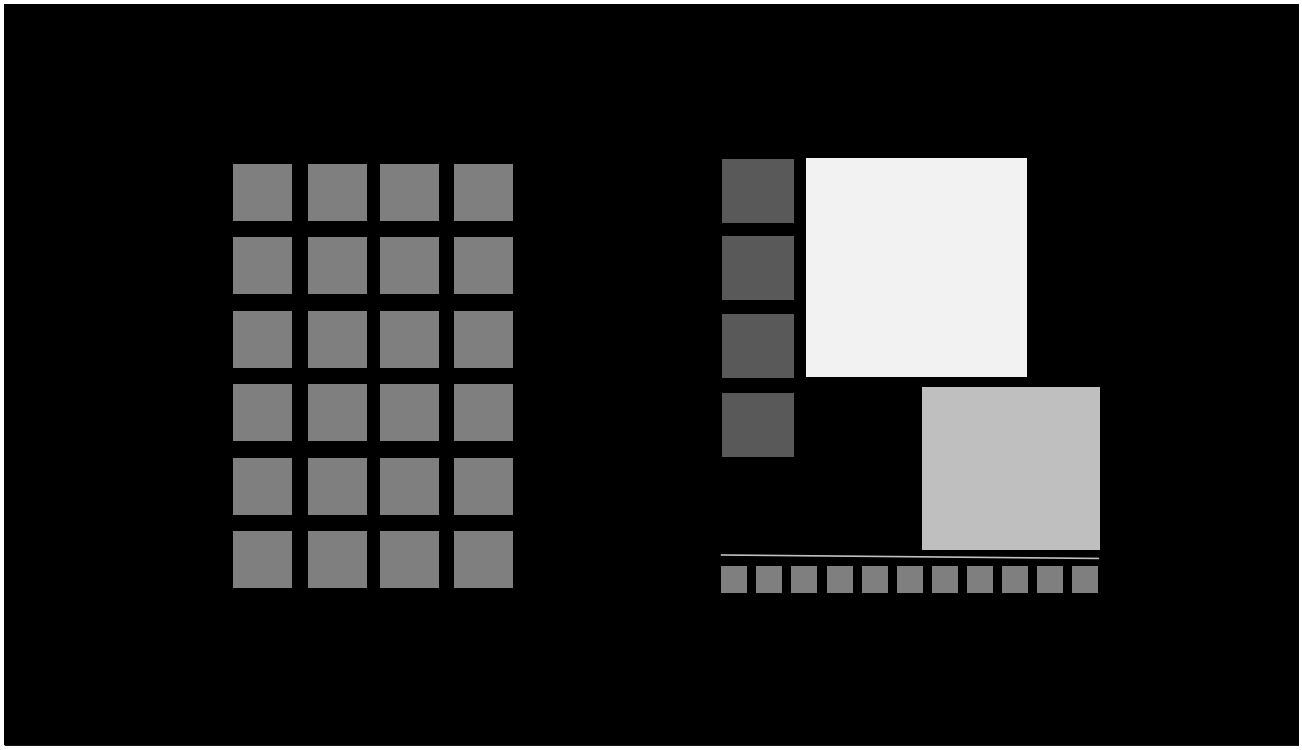
Affordance
Consistency
Constraints
Feedback
Mapping
▶ **Visibility**



Show/increase the (useful) information
Prioritise/sharpen the information

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9



10

CONCLUSION

- Do **not** conform to design guidelines
- Use the design guidelines and principles **as a tool to frame and think about your design**

➡ Notice our day-to-day UIs we interact with… think about how you can explain its good/poor usability by these guidelines/principles, how they could be designed better

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11

Design Exercise 1…

- Feedback & result shared this morning

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12

Design Exercise 1…

- Literature should be searched to support your design decisions
- Find similar studies/books/example products that do use your proposed approach, and reference them => support your design decisions with some evidences
- References: 2 from the project description ☹
- Use the concepts & terminologies that we learned in this module

13

Design Exercise 1…

- "button would be lit up", "Current floor to be displayed here", "highlight selected button", "Instruction here" …
- Page number and Table of Contents
- All entries in Reference section should be cited from the main body of report (also all figures and tables)
- All entries in Reference section need full bibliographic information

14

Design Exercise 1…

- Having emergency button and door buttons grouped… why?
- Current floor, up/down status
- Numeric pad UI: issues in usability?

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15

DX1 examples

We looked at 3 good examples and saw how/why they are good.

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16

Individual report

- Description available on Loop
- Deadline is the end of November, after the final group submission
- Check the rubric hints note
- No big workload

17